ReCon: Enhancing True Correspondence Discrimination through Relation Consistency for Robust Noisy Correspondence Learning

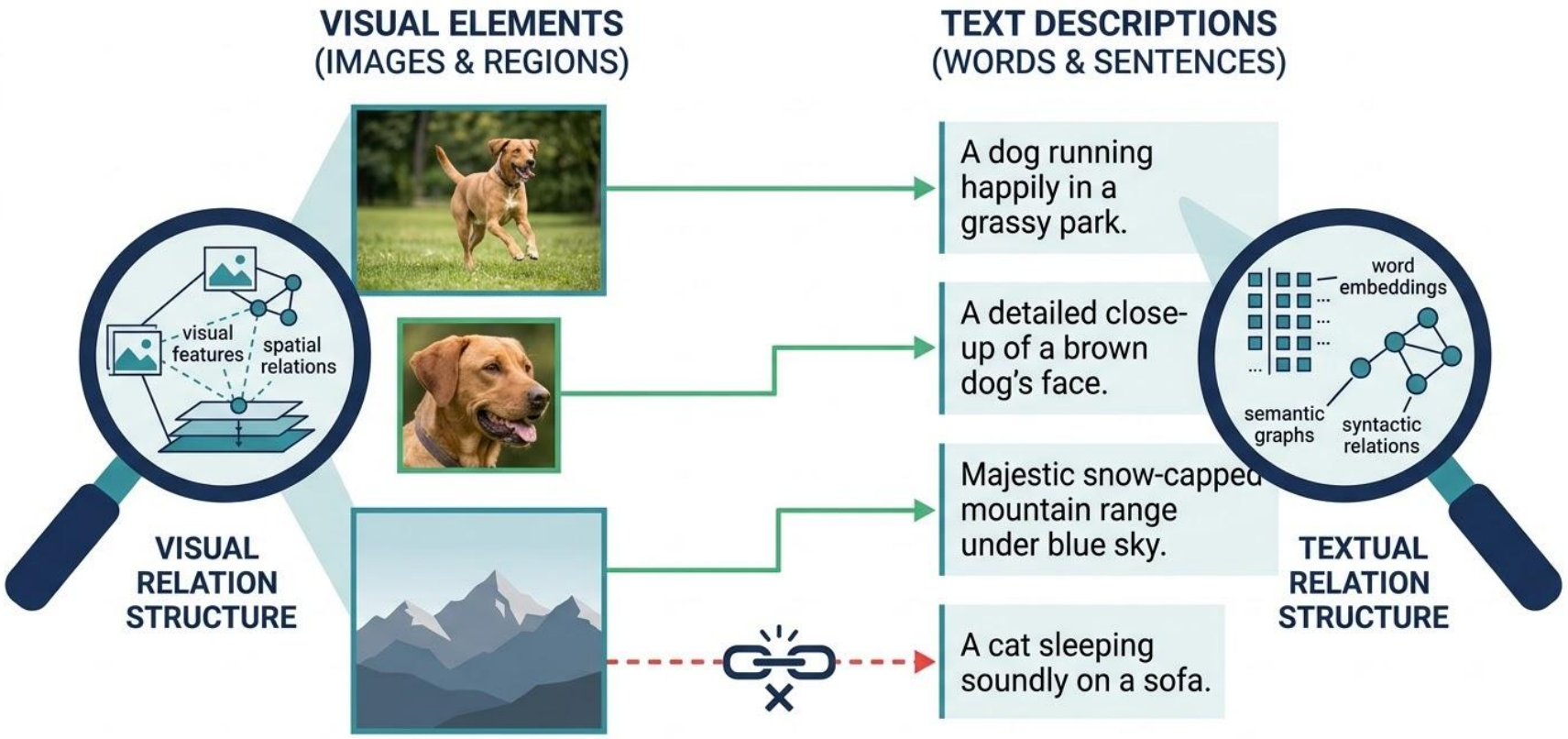
Quanxing Zha[1], Xin Liu[1,2*], Shu-Juan Peng[1], Yiu-ming Cheung[2], Xing Xu[3], Nannan Wang[4]
[1]Huaqiao University, [2]Hong Kong Baptist University
[3]University of Electronic Science and Technology of China, [4]Xidian University
quanxing.zha@gmail.com, xliu@hqu.edu.cn

Presenter: Cheng-Jun Kang

Institute of Data Science, National Cheng Kung University

Outline

- Authors
- Introduction
- Problem Definition
- Related Work
- Motivation
- Methodology
- Experiment
- Conclusion



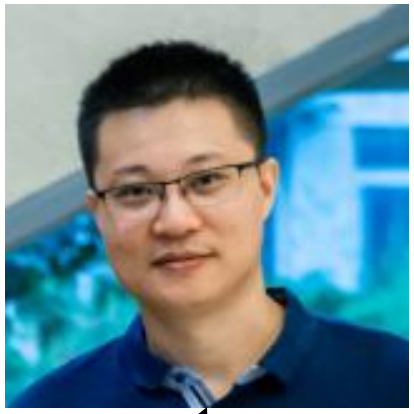
Authors

Media Analysis and Data Mining Group
- Department of Computer Science and Technology
- Huaqiao University, Xiamen 361021, China

华侨大学媒体分析和数据挖掘小组
Media Analysis and Data Mining

Quanxing Zha (查全興)
First Author
Interest:
- Noisy Correspondence Learning
- Multimodal Learning
- Multimodal Retrieval-augmented Generation

Shu-Juan Peng (彭淑娟)
Third Author
Interest:
- Fine-grained Image-text Retrieval
- Cross-modal hashing



Liu Xin (柳欣)
Second & Corresponding Author
Interest:
- Multimedia analysis
- Pattern recognition
- Data mining

Yiu-ming Cheung (張曉明)
Hong Kong Baptist University

Xing Xu (徐 行)
UESTC, China

NannanWang(王楠楠)
ISN State Key Laboratory, Xidian University

Introduction

Cross-Modal Retrieval(CMR)

aims to search the most relevant items across different modalities in response to query modality.

Image → Text Retrieval

Given an image, find the most matching text description.

Text → Image Retrieval

Given a text description, find the most matching image.

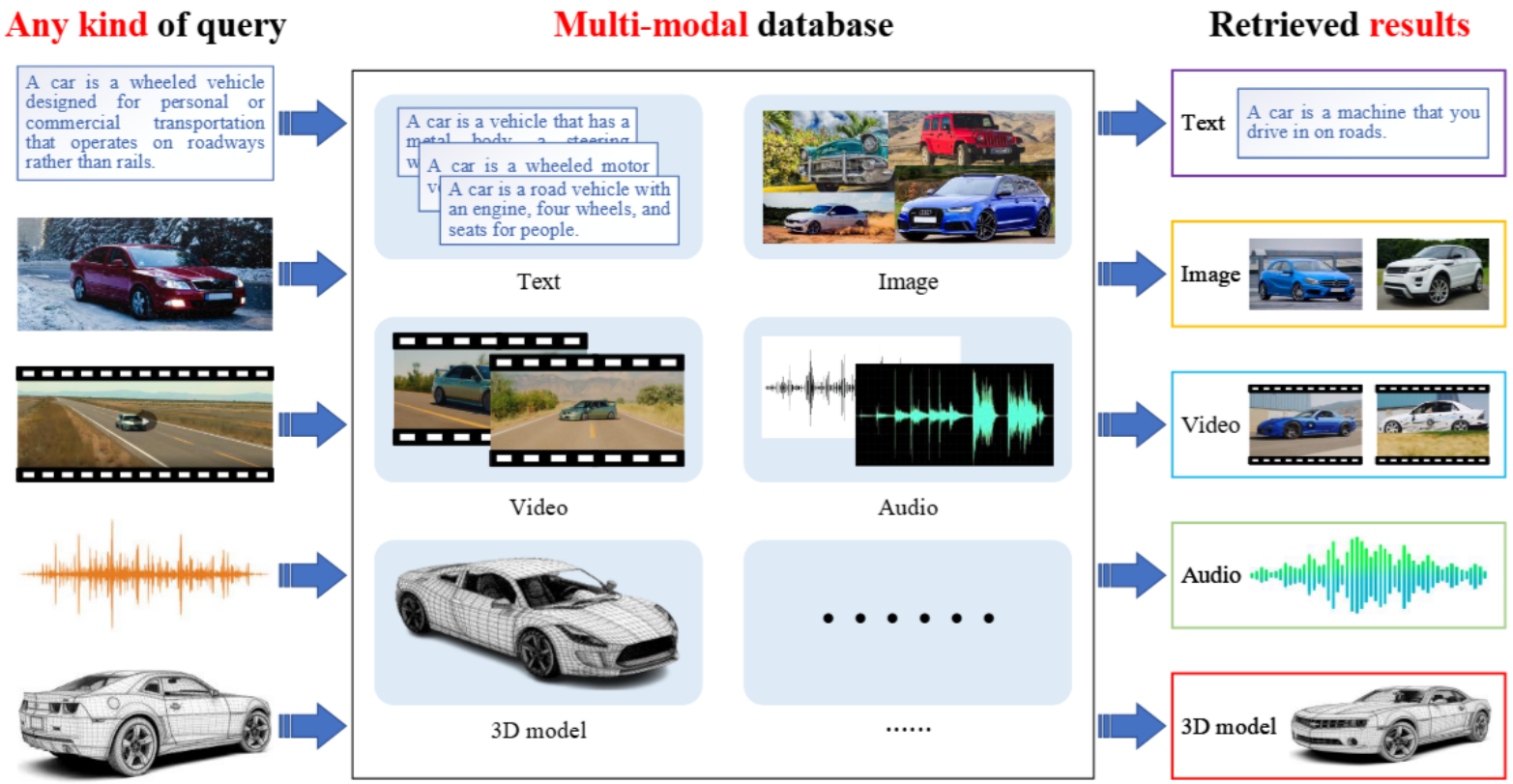


Figure 1. Illustration of cross-modal retrieval.[1]

[1] Wang, T., Li, F., Zhu, L., Li, J., Zhang, Z., & Shen, H. T. (2023). Cross-modal retrieval: a systematic review of methods and future directions. arXiv preprint arXiv:2308.14263.

Cross-Modal Retrieval Mainstream method

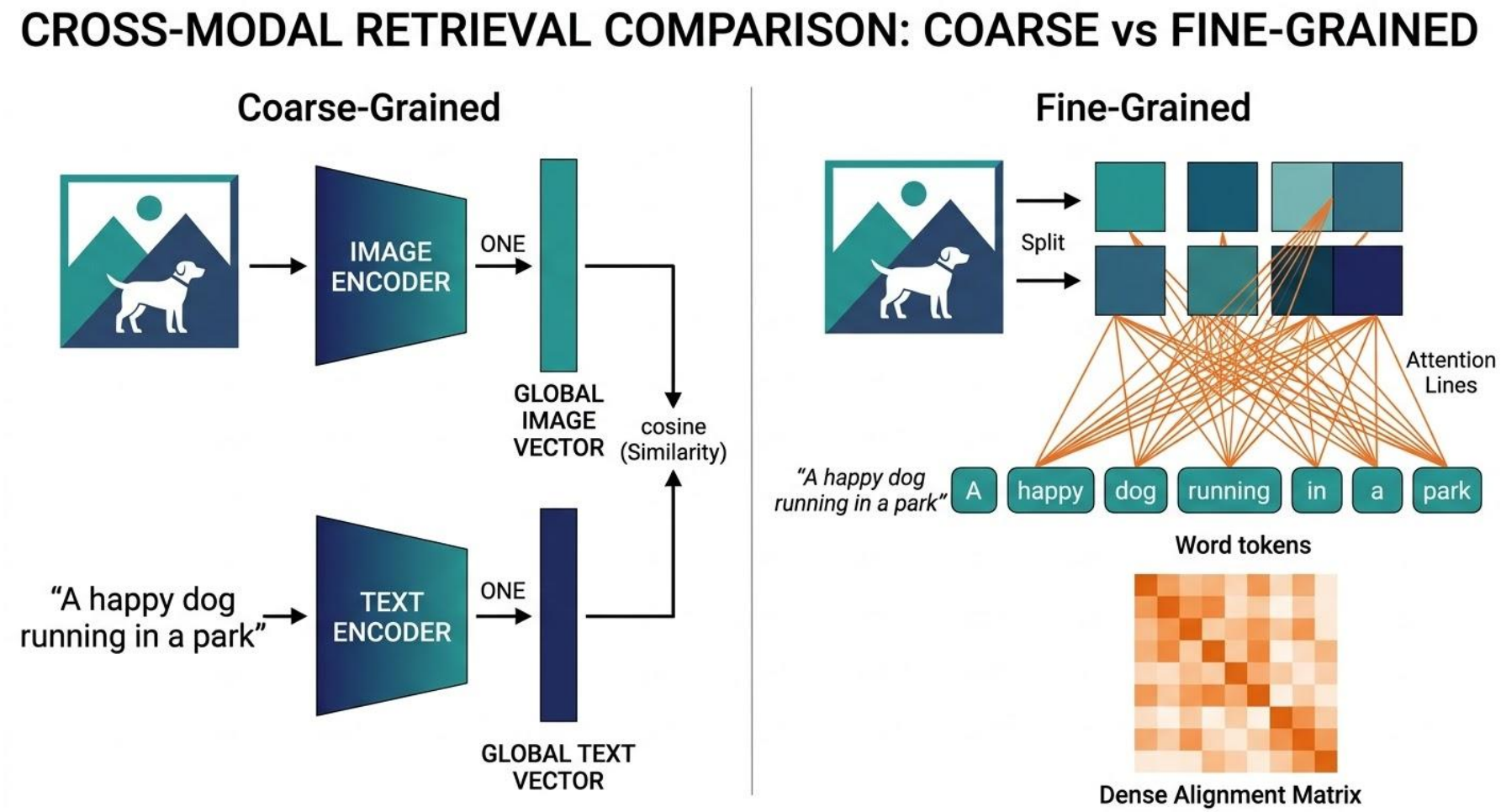


Figure 2. Coarse-Grained[2] vs. Fine-Grained [3]

[2] Li, K., Zhang, Y., Li, K., Li, Y., & Fu, Y. (2022). Image-text embedding learning via visual and textual semantic reasoning. IEEE transactions on pattern analysis and machine intelligence, 45(1), 641-656.
[3] Zhang, K., Mao, Z., Wang, Q., & Zhang, Y. (2022). Negative-aware attention framework for image-text matching. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 15661-15670).

Problem Definition

“ Typically, all visual-text pairs are assumed to be semantically associated with high similarity scores in the common representation space. However, due to the substantial costs of data collection and annotations, an unknown portion of mismatched pairs may be inadvertently labeled as matched ones.”

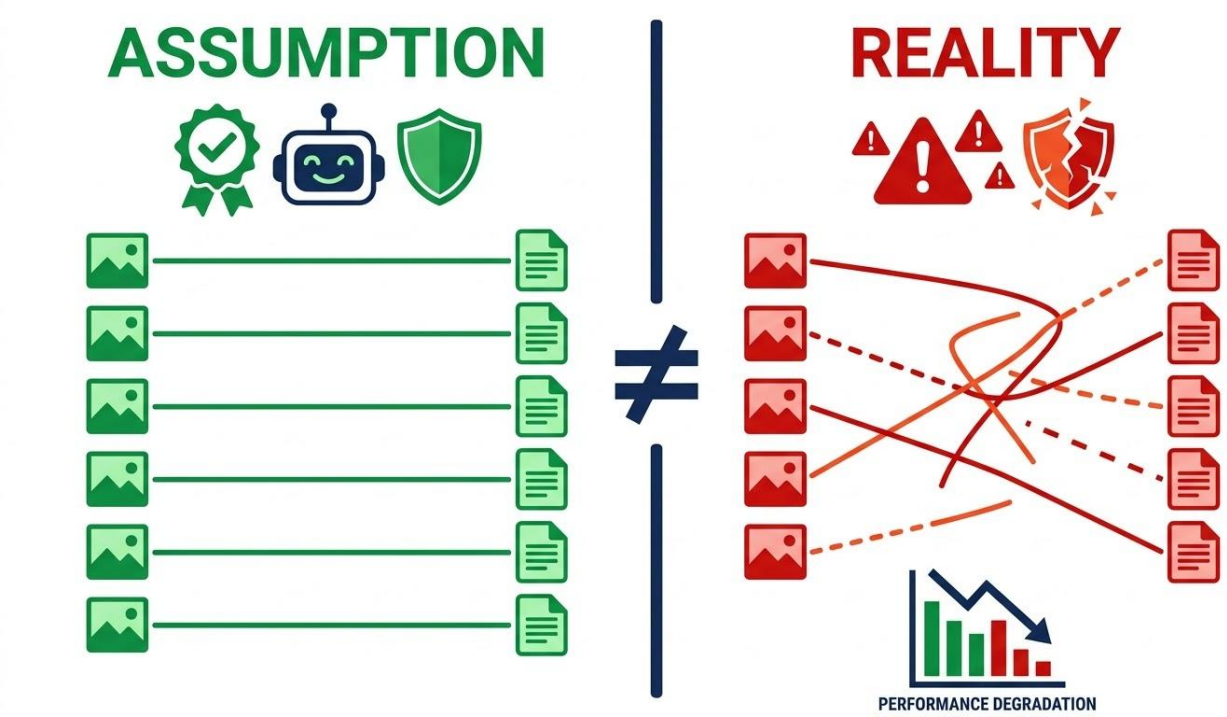


Figure 3. The ideal assumption vs. the noisy reality of cross-modal datasets.

Figure 4. Illustrations of the NC problem the dataset consists of both true and false positive pairs, while the ground truth is agnostic. NC will reduce the consistency of positives and mislead the optimization direction, thus degrading the performance of CMR models. [4]

[4] Ma, X., Yang, M., Li, Y., Hu, P., Lv, J., & Peng, X. (2024). Cross-modal retrieval with noisy correspondence via consistency refining and mining. IEEE transactions on image processing, 33, 2587-2598.

Related Work

1.

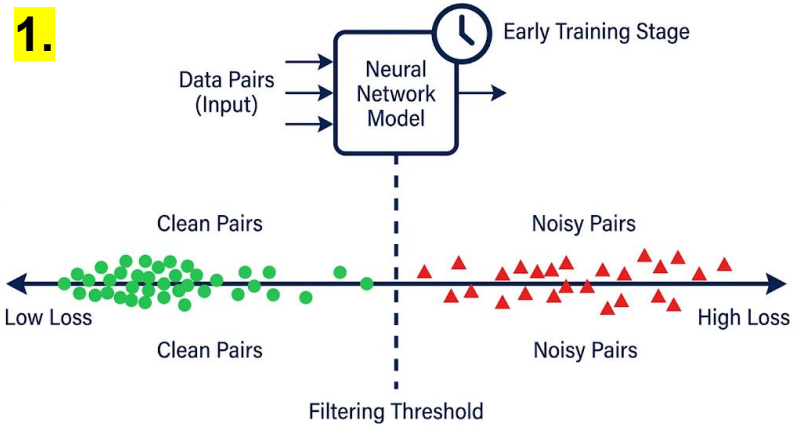


Figure 5. Small-Loss Criterion. [5]

2.

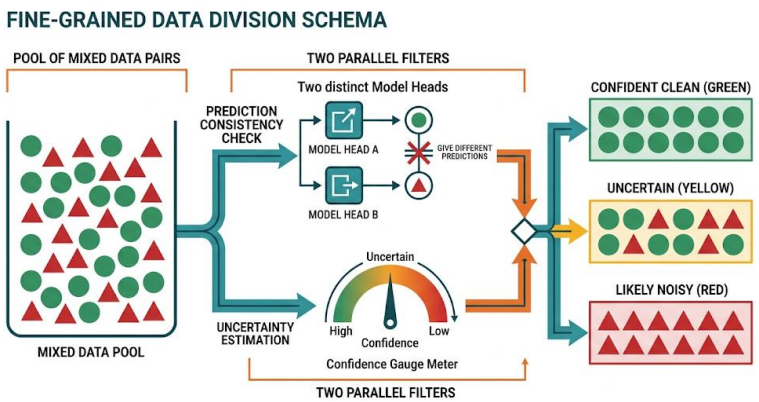


Figure 6. Fine-Grained Division.[6]

3.

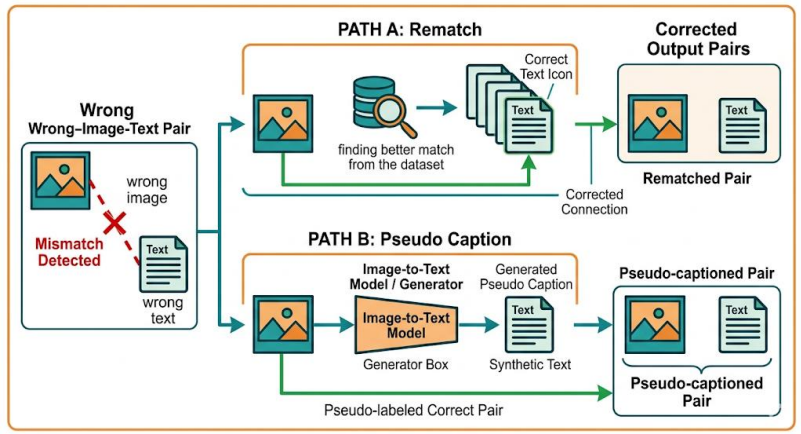


Figure 7. Alternative Strategies. [7]

4.

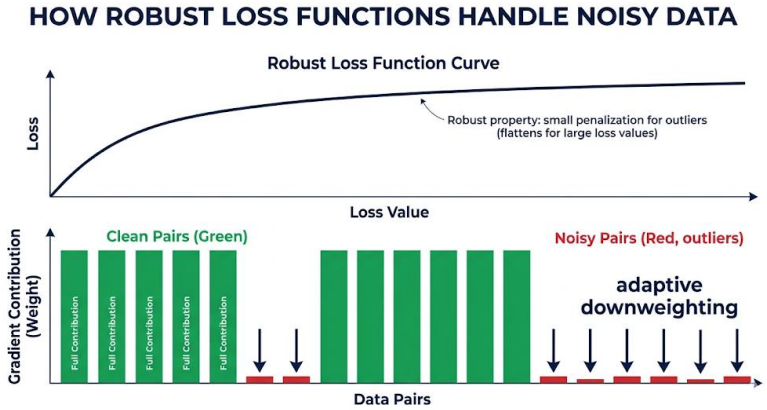


Figure 8. Robust Loss Functions.[8]

5.

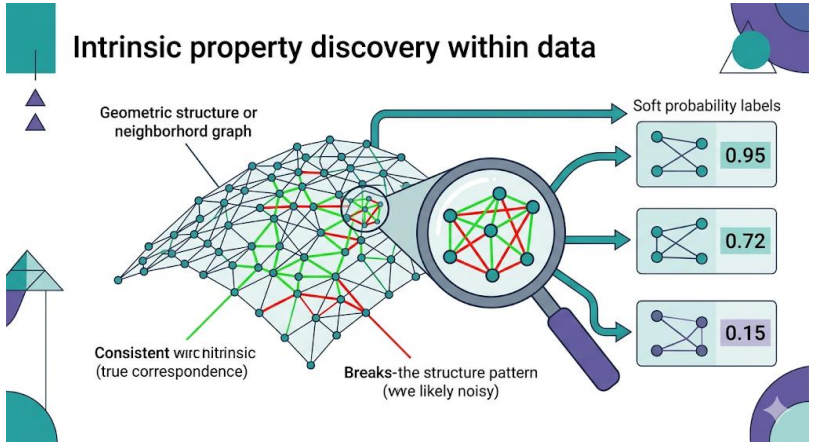


Figure 9. Intrinsic Properties. [9]

[5] Huang, Z., Niu, G., Liu, X., Ding, W., Xiao, X., Wu, H., & Peng, X. (2021). Learning with noisy correspondence for cross-modal matching. Advances in Neural Information Processing Systems, 34, 29406-29419.
[6] Feng, Z., Zeng, Z., Guo, C., Li, Z., & Hu, L. (2023). Learning from noisy correspondence with tri-partition for cross-modal matching. IEEE Transactions on Multimedia, 26, 3884-3896.
[7] Han, H., Zheng, Q., Dai, G., Luo, M., & Wang, J. (2024). Learning to rematch mismatched pairs for robust cross-modal retrieval. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 26679-26688).
[8] Qin, Y., Sun, Y., Peng, D., Zhou, J. T., Peng, X., & Hu, P. (2023). Cross-modal active complementary learning with self-refining correspondence. Advances in neural information processing systems, 36, 24829-24840.
[9] Zhao, Z., Chen, M., Dai, T., Yao, J., Han, B., Zhang, Y., & Wang, Y. (2024). Mitigating noisy correspondence by geometrical structure consistency learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 27381-27390).

Motivation

Relation consistency enhances retrieval

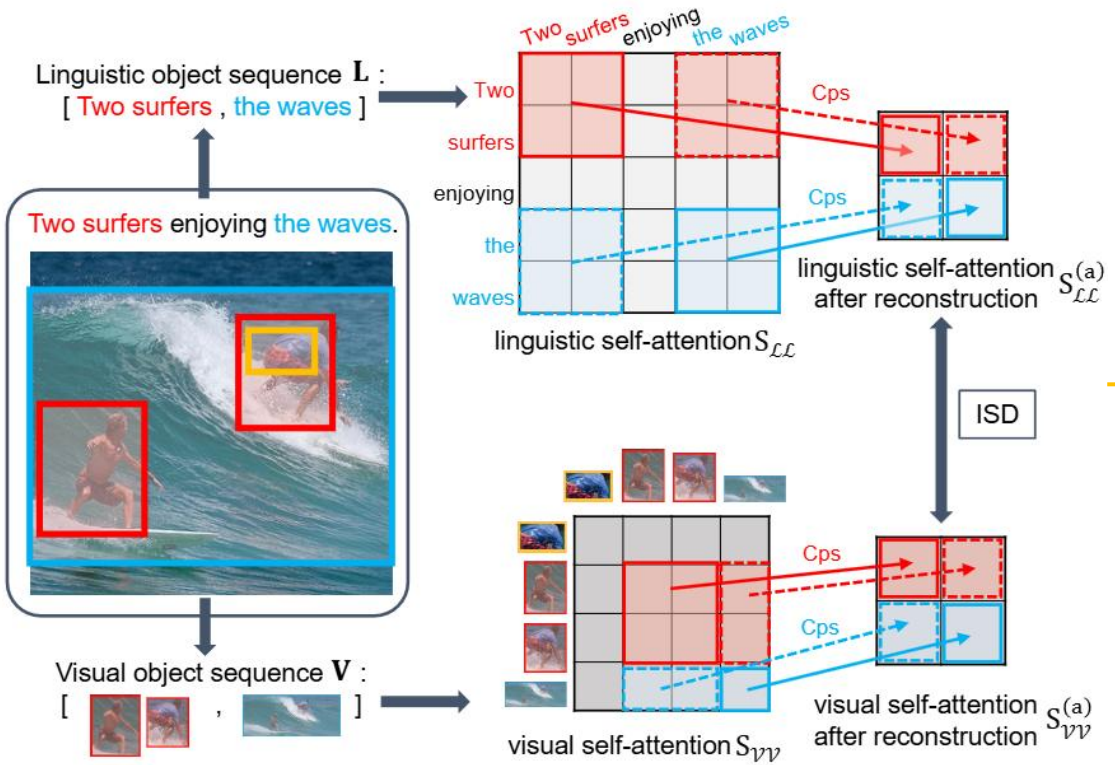


Figure 10. An example of calculating Intra-modal Self attention Distance for a matched image-text pair.[10]

Relation inconsistency detects noise

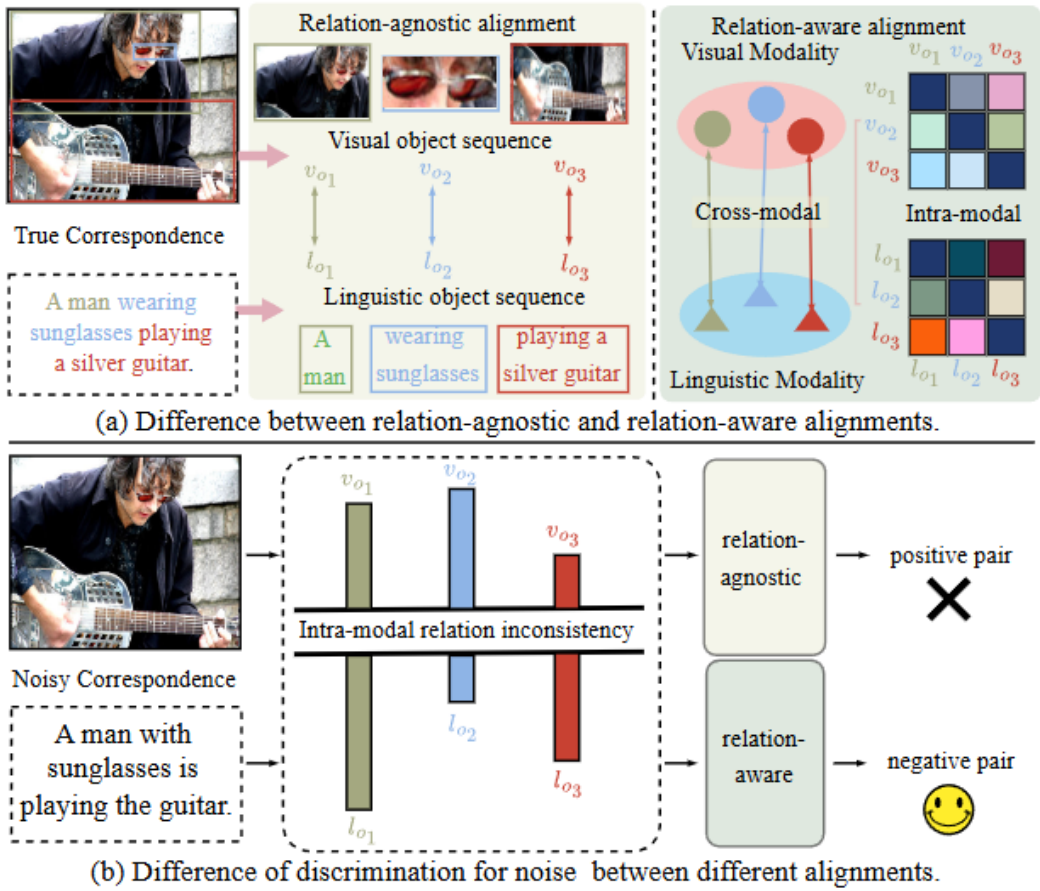


Figure 11. Illustration of relation discrepancy. The relation-aware alignment correctly identifies mismatched pair as negatives, while relation-agnostic alignment fails to detect such inconsistency.

[10] Ren, S., Lin, J., Zhao, G., Men, R., Yang, A., Zhou, J., ... & Yang, H. (2021, August). Learning relation alignment for calibrated cross-modal retrieval. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (pp. 514-524).

Methodology

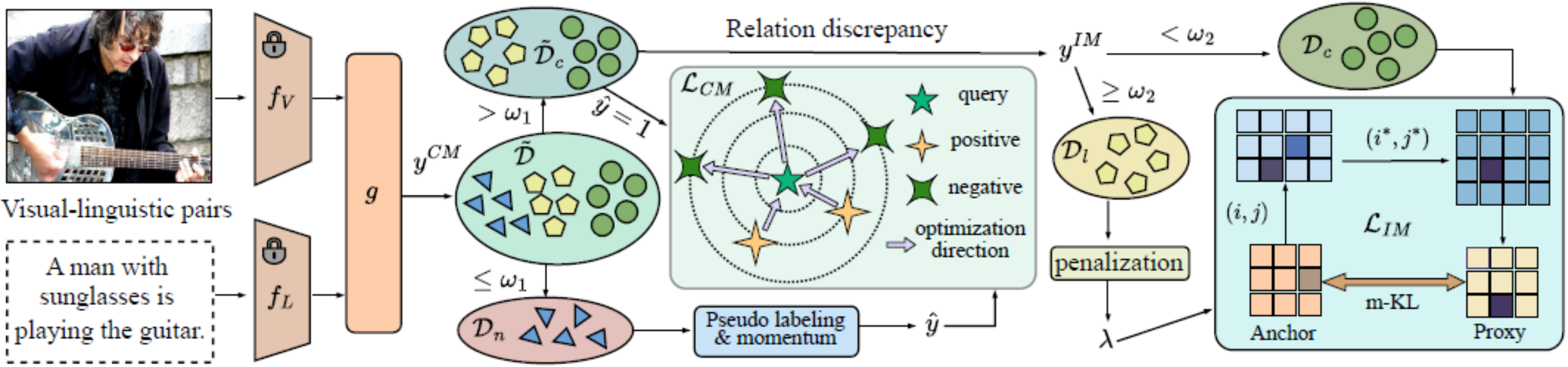


Figure 12. The schematic pipeline of the proposed ReCon learning framework.

Methodology

$$\mathcal{L}_c = \xi \mathcal{L}_{\mathcal{CM}} + \mathcal{L}_{\mathcal{IM}}$$

$$\mathcal{L}_\ell = \xi \mathcal{L}_{\mathcal{CM}} + \frac{1}{\lambda} \mathcal{L}_{\mathcal{IM}}$$

$$\mathcal{L}_n = \hat{y}_i \mathcal{L}_{\mathcal{CM}}$$

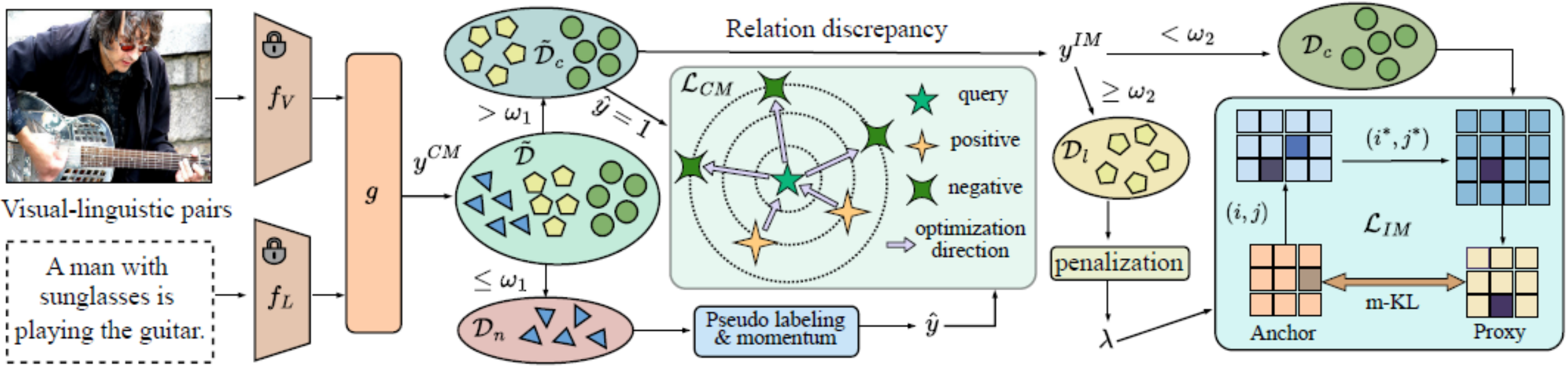
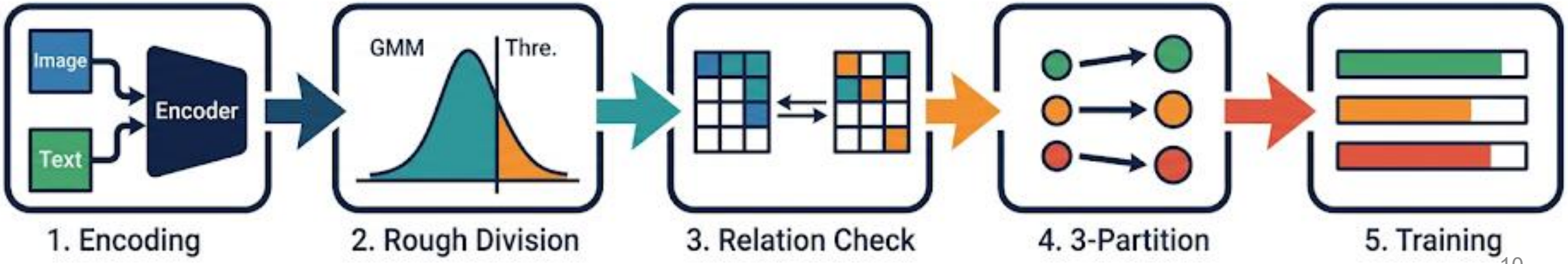


Figure 12. The schematic pipeline of the proposed ReCon learning framework.



Experiment (Dataset)

Flickr30k[11]

Gray haired man in black suit and yellow tie working in a financial environment.

A butcher cutting an animal to sell.

MS-COCO[12]

The man at bat readies to swing at the pitch while the umpire looks on.

A horse carrying a large load of hay and two people sitting on it.

Conceptual Captions(CC)[13]

Conceptual Captions: A worker helps to clear the debris.

Conceptual Captions: pop artist performs at the festival in a city.

[11] Young, P., Lai, A., Hodosh, M., & Hockenmaier, J. (2014). From image descriptions to visual denotations: New similarity metrics for semantic inference over event descriptions. Transactions of the association for computational linguistics, 2, 67-78.

[12] Chen, X., Fang, H., Lin, T. Y., Vedantam, R., Gupta, S., Dollár, P., & Zitnick, C. L. (2015). Microsoft coco captions: Data collection and evaluation server. arXiv preprint arXiv:1504.00325.

[13] Sharma, P., Ding, N., Goodman, S., & Soricut, R. (2018, July). Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. In Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers) (pp. 2556-2565).

Experiment

$$\mathbf{R@K} = \frac{\text{Number of queries whose ground-truth match appears in the top } K \text{ results}}{\text{Total number of queries}}$$

for $K \in \{1, 5, 10\}$

$$\mathbf{rSum} = \mathbf{R@1}_{I\to T} + \mathbf{R@5}_{I\to T} + \mathbf{R@10}_{I\to T} + \mathbf{R@1}_{T\to I} + \mathbf{R@5}_{T\to I} + \mathbf{R@10}_{T\to I}$$

where Max = 600.

Noise Ratio	Methods	Flickr30K							MS-COCO 1K						
		Image to Text			Text to Image				Image to Text			Text to Image			
		R@1	R@5	R@10	R@1	R@5	R@10	rSum	R@1	R@5	R@10	R@1	R@5	R@10	rSum
20%	SCAN (ECCV'18)	59.1	83.4	90.4	36.6	67.0	77.5	414.0	66.2	91.0	96.4	45.0	80.2	89.3	468.1
	NCR (NIPS'21)	73.5	93.2	96.6	56.9	82.4	88.5	491.1	76.6	95.6	98.2	60.8	88.8	95.0	515.0
	DECL (ACM MM'22)	77.5	93.8	97.0	56.1	81.8	88.5	494.7	77.5	95.9	98.4	61.7	89.3	95.4	518.2
	MSCN (CVPR'23)	77.4	94.9	97.6	59.6	83.2	89.2	501.9	78.1	**97.2**	98.8	64.3	90.4	95.8	524.6
	BiCro (CVPR'23)	78.1	94.4	97.5	60.4	84.4	89.9	504.7	78.8	96.1	98.6	63.7	90.3	95.7	523.2
	RCL (TPAMI'23)	75.9	94.5	97.3	57.9	82.6	88.6	496.8	78.9	96.0	98.4	62.8	89.9	95.4	521.4
	CRCL (NIPS'23)	78.9	94.8	**97.9**	58.7	83.0	89.2	502.5	77.8	96.1	98.5	63.4	90.3	95.9	522.0
	SREM (AAAI'24)	79.5	94.2	**97.9**	61.2	84.8	90.2	507.8	78.5	96.8	98.8	63.8	90.4	95.8	524.1
	PC^2 (ACM MM'24)	78.7	94.9	96.9	59.8	83.9	89.6	503.8	77.8	95.7	98.4	62.8	89.7	95.3	519.7
	L2RM (CVPR'24)	77.9	95.2	97.8	59.8	83.6	89.5	503.8	80.2	96.3	98.5	64.2	90.1	95.4	524.7
	ESC (CVPR'24)	79.0	94.8	97.5	59.1	83.8	89.1	503.3	79.2	97.0	**99.1**	64.8	90.7	**96.0**	526.8
	GSC (CVPR'24)	78.3	94.6	97.8	60.1	84.5	90.5	505.8	79.5	96.4	98.9	64.4	90.6	95.9	525.7
	ReCon	**80.3**	**95.3**	97.8	**61.6**	**85.5**	**91.3**	**511.8**	**80.9**	96.6	98.8	**65.2**	**91.0**	**96.0**	**528.6**
40%	SCAN (ECCV'18)	29.9	60.5	72.5	16.4	38.5	48.6	266.4	30.1	65.2	79.2	18.9	51.1	69.9	314.4
	NCR (NIPS'21)	75.3	92.1	95.2	56.2	80.6	87.4	486.8	76.5	95.0	98.2	60.7	88.5	95.0	513.9
	DECL (ACM MM'22)	72.7	92.3	95.4	53.4	79.4	86.4	479.6	75.6	95.5	98.3	59.5	88.3	94.8	512.0
	MSCN (CVPR'23)	74.4	**94.4**	96.9	57.2	81.7	87.6	492.2	74.8	94.9	98.0	60.3	88.5	94.4	510.9
	BiCro (CVPR'23)	74.6	92.7	96.2	55.5	81.1	87.4	487.5	77.0	95.9	98.3	61.8	89.2	94.9	517.1
	RCL (TPAMI'23)	72.7	92.7	96.1	54.8	80.0	87.1	483.4	77.0	95.5	98.3	61.2	88.5	94.8	515.3
	CRCL (NIPS'23)	74.1	92.6	96.9	55.5	80.9	87.6	487.6	76.6	95.6	98.5	62.3	89.7	95.4	518.1
	SREM (AAAI'24)	76.5	93.9	96.3	57.5	82.7	88.5	495.4	77.2	96.0	98.5	62.1	89.3	95.3	518.4
	PC^2 (ACM MM'24)	75.8	93.5	96.9	57.5	81.9	88.2	493.8	77.4	95.8	98.4	62.1	89.4	95.1	518.2
	L2RM (CVPR'24)	75.8	93.2	96.9	56.3	81.0	87.3	490.5	77.5	95.8	98.4	62.0	89.1	94.9	517.7
	ESC (CVPR'24)	76.1	93.1	96.4	56.0	80.8	87.2	489.6	78.6	**96.6**	**99.0**	63.2	**90.6**	**95.9**	523.9
	GSC (CVPR'24)	76.5	94.1	**97.6**	57.5	82.7	88.9	497.3	78.2	95.9	98.2	62.5	89.7	95.4	519.9
	ReCon	**79.4**	94.3	**97.6**	**59.9**	**83.9**	**90.1**	**505.2**	**79.9**	96.2	98.6	**63.5**	90.5	**95.9**	**524.5**
60%	SCAN (ECCV'18)	16.9	39.3	53.9	2.8	7.4	11.4	131.7	27.8	59.8	74.8	16.8	47.8	66.4	293.4
	NCR (NIPS'21)	68.7	89.9	95.5	52.0	77.6	84.9	468.6	72.7	94.0	97.6	57.9	87.0	94.1	503.3
	DECL (ACM MM'22)	65.2	88.4	94.0	46.8	74.0	82.2	450.6	73.0	94.2	97.9	57.0	86.6	93.8	502.5
	MSCN (CVPR'23)	70.4	91.0	94.9	53.4	77.8	84.1	471.6	74.4	95.1	97.9	59.2	87.1	92.8	506.5
	BiCro (CVPR'23)	67.6	90.8	94.4	51.2	77.6	84.7	466.3	73.9	94.4	97.8	58.3	87.2	93.9	505.5
	RCL (TPAMI'23)	67.7	89.1	93.6	48.0	74.9	83.3	456.6	74.0	94.3	97.5	57.6	86.4	93.5	503.3
	CRCL (NIPS'23)	70.4	90.4	94.9	52.6	78.1	85.1	471.5	75.2	94.9	98.0	60.1	88.5	94.8	511.5
	SREM (AAAI'24)	71.0	92.1	96.1	54.0	80.1	87.0	480.3	74.5	94.5	97.9	58.7	87.5	93.9	506.9
	PC^2 (ACM MM'24)	70.8	90.3	94.4	53.1	79.0	85.9	473.5	74.2	94.4	97.8	58.9	87.5	93.8	506.6
	L2RM (CVPR'24)	70.0	90.8	95.4	51.3	76.4	83.7	467.6	75.4	94.7	97.9	59.2	87.4	93.8	508.4
	ESC (CVPR'24)	72.6	90.9	94.6	53.0	78.6	85.3	475.0	**77.2**	95.1	98.1	61.1	88.6	94.9	515.0
	GSC (CVPR'24)	70.8	91.1	95.9	53.6	79.8	86.8	478.0	75.6	95.1	98.0	60.0	88.3	94.6	511.7
	ReCon	**74.3**	**93.6**	**96.6**	**55.7**	**81.6**	**88.1**	**489.9**	**77.2**	**95.9**	**98.4**	**61.8**	**89.3**	**95.2**	**517.8**

Table 1. Cross-modal retrieval performance comparison under synthetic noise rates of 20%, 40%, and 60% on Flickr30K and MS-COCO1K. The best and the second best results are respectively marked by **bold** and underline.

Experiment

Methods	Image to Text			Text to Image			
	R@1	R@5	R@10	R@1	R@5	R@10	rSum
SCAN	30.5	55.3	65.3	26.9	53.0	64.7	295.7
NCR	39.5	64.5	73.5	40.3	64.6	73.2	355.6
DECL	39.0	66.1	75.5	40.7	66.3	76.7	364.3
MSCN	40.1	65.7	76.6	40.6	67.4	76.3	366.7
BiCro	40.8	67.2	76.1	42.1	67.6	76.4	370.2
RCL	41.7	66.0	73.6	41.6	66.4	75.1	364.4
CRCL	41.8	67.4	76.5	41.6	68.0	**78.4**	373.7
SREM	40.9	67.5	77.1	41.5	68.2	77.0	372.2
PC^2	39.3	66.4	75.4	39.8	66.4	76.8	364.1
L2RM	43.0	67.5	75.7	42.8	68.0	77.2	374.2
ESC	42.8	67.3	76.9	44.8	68.2	75.9	375.9
GSC	42.1	68.4	77.7	42.2	67.6	77.1	375.1
ReCon	**43.1**	**68.7**	**78.1**	**44.9**	**68.3**	77.4	**380.5**

Table 2. Comparisons with real-world NCs on CC152K. The **Best** and second-best results are respectively marked in each column.

Components			Image to Text			Text to Image			
Tru.	$\mathcal{L}_{IM}$	λ	R@1	R@5	R@10	R@1	R@5	R@10	rSum
✓	✓	✓	**79.4**	**94.3**	**97.6**	**59.9**	**83.9**	**90.1**	**505.2**
✓	✓		77.3	94.1	97.3	58.7	83.3	89.5	500.2
	✓	✓	77.2	**94.3**	97.2	57.9	83.1	89.3	499.1
	✓		77.0	94.1	97.0	57.6	82.8	89.0	497.5
✓			74.1	93.2	96.7	57.4	83.1	88.9	493.3

Table 3. Ablation studies on Flick30K with 40% noise with different components in ReCon. The **best** results are marked in bold.

hipster man walking in the streets.

making a serious point during press conference.

Figure 13. Some detected noisy examples by ReCon on CC152K..

Conclusion

"The main motivation of ReCon is to *enhance the discriminability of models* for true correspondences in noisy multimodal dataset."

- Leverage dual constraints (cross- & intra-modal) to effectively identify "Hard NCs" that relation-agnostic methods miss.
- Achieves SOTA performance, showing high robustness even under severe noise (e.g., 60%) and on real-world datasets (CC152K).

Discussion / My View

Strengths: Dual-constraint is intuitive and effective; specifically good at filtering “Hard NCs” by design. Achieves SOTA performance, showing high robustness even under severe noise (e.g., 60%) and on real-world datasets (CC152K).

Limitations:

- **Outdated Backbone:** Relies on SGRAF. Effectiveness on modern VLMs (e.g., BLIP-2, CLIP variants) remains unverified.
- **Information Loss in Proxy:** Uses argmax (hard assignment) for proxy construction. Soft attention might preserve richer context.
- **Rigid Thresholds:** $\omega_1, \omega_2 = 0.5$ are fixed. Adaptive thresholds might be better across different noise rates.

Future Work:

- Apply to VQA, Visual Grounding.
- Explore "Soft Proxy" construction.
- Scale up for massive web-crawled noisy pretraining.

Thanks for Your Attention